Notes on the Ising perceptron at nonzero margin

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Abstract

We study the replica-symmetric system behind the Ising perceptron capacity at general margin κ. Two elementary identities make the general-κ system as well behaved as the zero-margin one: the free energy is stationary in (q, ψ) exactly at the fixed point, so the threshold $\alpha_\star(\kappa)$ is the unique root of a strictly decreasing function; and dR/dq has a one-term integral representation with positive integrand, so the fixed-point map is monotone at every κ. We compute the capacity candidate curve and the numerical content of Huang's conditions on $\kappa \in [-2, 1.5]$, and we certify, in ball arithmetic, the fixed-point condition together with the AMP and local-concavity conditions on the strip $\kappa \in [-0.65, 0.19]$ (the last 0.09 beyond the reach of the rectangle product bound, on Nakajima's certified premise), with $\alpha_\star(\kappa)$ located to $2 \cdot 10^{-6}$ at five margins and the local maximality of the first-moment functional at the distinguished point certified at six margins spanning the strip. The global part of the two-variable condition remains open at $\kappa \neq 0$; its numerical landscape is essentially independent of κ on the range scanned.

1 The system

Let φ and Ψ be the standard Gaussian density and upper tail, $M = \varphi/\Psi$ the inverse Mills ratio, and $Z \sim N(0,1)$. Fix $\kappa \in \mathbb{R}$ and set

$$u(z; q, \kappa) = \frac{\kappa - \sqrt{q}\, z}{\sqrt{1-q}}, \qquad 0 < q < 1.$$

Following Ding–Sun and Huang, define

$$P(\psi) = \mathbb{E}\big[\tanh^2(\sqrt{\psi}\, Z)\big], \qquad R(q, \alpha) = \frac{\alpha}{1-q}\, \mathbb{E}\big[M(u(Z))^2\big], \tag{1}$$

$$G(\alpha, q, \psi) = -\frac{\psi(1-q)}{2} + \mathbb{E}\log\big(2\cosh(\sqrt{\psi}\, Z)\big) + \alpha\, \mathbb{E}\log\Psi\big(u(Z)\big). \tag{2}$$

The replica-symmetric prediction is that the storage threshold is $\alpha_\star(\kappa)$, the zero of $\alpha \mapsto G_\star(\alpha) = G(\alpha, q_\star(\alpha), \psi_\star(\alpha))$ along the fixed-point branch $q = P(\psi)$, $\psi = R(q, \alpha)$. At $\kappa = 0$ this is the Krauth–Mézard value $\alpha_\star = 0.833078599\ldots$, now verified without numerical hypotheses; Huang's general-κ theorem upper-bounds the capacity by $\alpha_\star(\kappa)$ for every κ, conditional on four numerical conditions of which only the $\kappa = 0$ instances had been checked.

2 Two identities

Lemma 1 (Stationarity)**.** *For all $\alpha > 0$, $\kappa \in \mathbb{R}$, $\psi > 0$ and $0 < q < 1$,*

$$\frac{\partial G}{\partial \psi} = \frac{q - P(\psi)}{2}, \qquad \frac{\partial G}{\partial q} = \frac{\psi - R(q,\alpha)}{2}.$$

Proof. The first identity is Gaussian integration by parts on the log cosh term. For the second, differentiating u in q gives

$$u_q = \frac{1}{2\sqrt{1-q}}\Big(-\frac{z}{\sqrt{q}} + \frac{u}{\sqrt{1-q}}\Big), \qquad \frac{\partial}{\partial q}\,\mathbb{E}\log\Psi(u) = -\mathbb{E}\big[M(u)\,u_q\big].$$

Stein's lemma applied to the z part, $\mathbb{E}[zM(u)] = -\sqrt{q/(1-q)}\;\mathbb{E}[M'(u)]$, together with the Mills identity $M'(u) = M(u)^2 - uM(u)$, collapses the two pieces:

$$\mathbb{E}[M(u)\,u_q] = \frac{\mathbb{E}[M'(u)] + \mathbb{E}[u\,M(u)]}{2(1-q)} = \frac{\mathbb{E}[M(u)^2]}{2(1-q)} = \frac{R(q,\alpha)}{2\alpha}.$$

Hence $\partial_q G = \psi/2 - \alpha \cdot R/(2\alpha) = (\psi - R)/2$. □

Both derivatives vanish exactly on the fixed-point branch, so the envelope theorem gives $G'_\star(\alpha) = \mathbb{E}\log\Psi(u) < 0$ strictly, for every κ: the root $\alpha_\star(\kappa)$ is unique on the branch. This is the general-κ form of the monotonicity clause in the fixed-point condition.

Lemma 2 (Derivative of R)**.** *For all $\alpha > 0$, $\kappa \in \mathbb{R}$ and $0 < q < 1$,*

$$\frac{\partial R}{\partial q} = \frac{\alpha}{(1-q)^2}\,\mathbb{E}\Big[M'(u)^2 + 2\,M(u)^2 M'(u)\Big] \;>\; 0.$$

Proof. Differentiating (1) in q produces $R/(1-q) + \frac{\alpha}{1-q}\mathbb{E}[2MM'u_q]$. Splitting u_q as in Lemma 1 and applying Stein's lemma to $\mathbb{E}[z\,MM'(u)]$ (with $(MM')' = M'^2 + MM''$ and $M'' = M'(M-u) + MM' - M$) turns the cross term into

$$\mathbb{E}[2MM'u_q] = \frac{\mathbb{E}[M'^2 + 2M^2M' - M^2]}{1-q},$$

and the $-\mathbb{E}[M^2]/(1-q)^2$ piece cancels $R/(1-q)$ exactly. Positivity is immediate since $M' \in (0,1)$. □

Lemma 2 replaces Ding–Sun's Lemma 7.2 (monotonicity of R, stated at $\kappa = 0$) at every margin, and its one-term form is what makes a tight interval enclosure of dR/dq possible. We checked both identities against high-precision finite differences (agreement to 10^{-15}).

3 The capacity candidate curve

Warm-started continuation from $\kappa = 0$ (cold solves above the fold escape to the frozen branch $q \to 1$) gives the curve of Figure 1 on $\kappa \in [-2.0, 1.5]$, with the fixed point, the envelope derivative, and the condition values recorded at every grid point. Selected values:

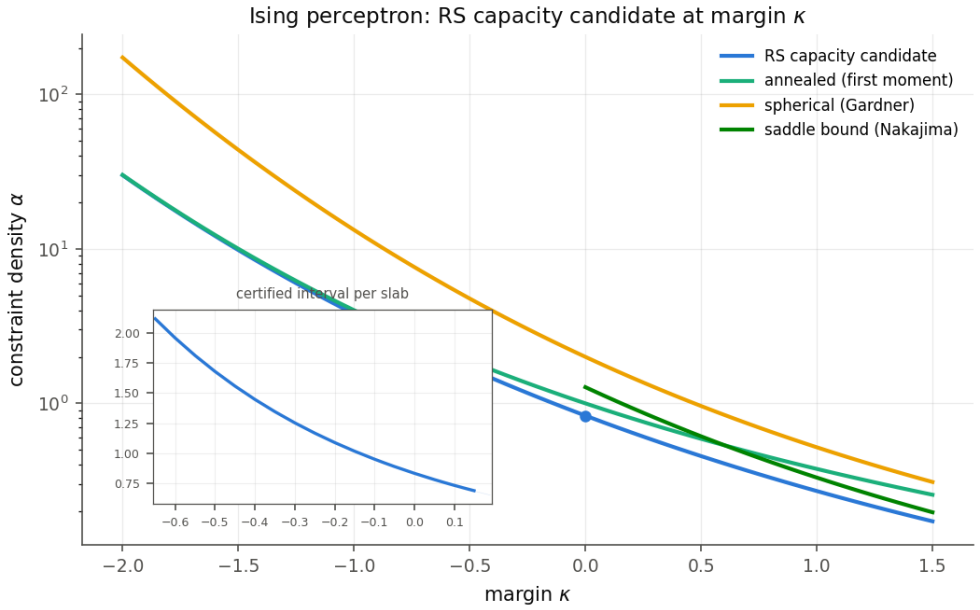


Figure 1: The replica-symmetric capacity candidate $\alpha_\star(\kappa)$ (continuation from the certified value at $\kappa = 0$), the annealed bound $\log 2/(-\log \Psi(\kappa))$, the Gardner spherical capacity, and Nakajima's saddle-existence bound $\alpha_c(\kappa) = 2/(\pi\, \mathbb{E}[(\kappa - Z)_+^2])$ for $\kappa \geq 0$.

κ	$\alpha_\star$	$q_\star$	$(P \circ R)'$	amp	λ_0
-2.0	29.9921	0.11863	0.387	0.440	-0.147
-1.0	3.79463	0.36519	0.583	0.607	-0.099
-0.5	1.67884	0.47377	0.666	0.589	-0.132
0	0.83308	0.56395	0.744	0.544	-0.191
0.5	0.45566	0.63841	0.810	0.489	-0.271
1.0	0.27070	0.69988	0.863	0.433	-0.359
1.5	0.17245	0.75049	0.903	0.381	-0.442

All three conditions hold at every point computed (Figure 2). The contraction is the only quantity trending toward failure, and only slowly; λ_0 approaches zero nearest $\kappa \approx -1.3$ (about -0.094) and then strengthens again. On the positive branch $\alpha_\star(\kappa)$ stays below Nakajima's saddle-existence bound at every grid point (worst ratio 0.874 at $\kappa = 1.5$), so saddle uniqueness there is covered by his theorem independently of our certificates. As $\kappa \to -2$ the candidate approaches the annealed bound from below, as expected where the constraints trivialize.

4 The two-variable function at general κ

Huang's global condition asks that $\mathcal{S}_\star(\lambda_1, \lambda_2) \leq 0$ everywhere, with equality at $(1, 0)$. Writing his functional at general κ (the margin enters both as an additive shift in the $\log \Psi$ argument and inside the planted cavity law $N(z) = M(u(z))/\sqrt{1 - q_0}$), the anchor $\mathcal{S}_\star(1, 0) = 0$ holds along the whole computed branch (to $2 \cdot 10^{-14}$ in double precision), with optimal tilt $s =$

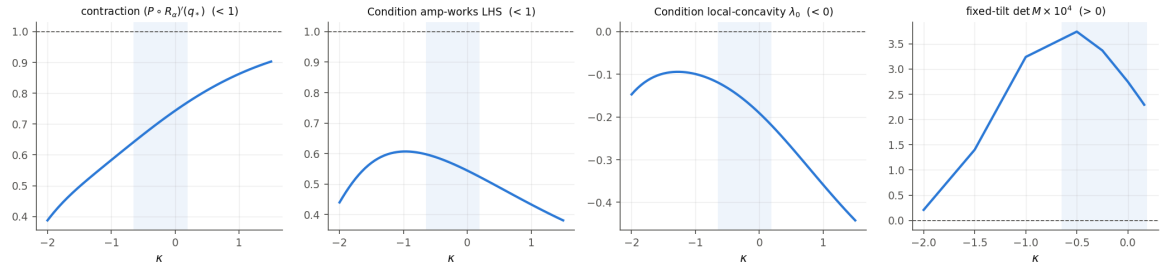


Figure 2: The numerical content of Huang's conditions along the branch: the contraction $(P \circ R_\alpha)'(q_\star)$, the AMP condition value, the local-concavity minimum λ_0, and the determinant of the fixed-tilt Hessian at the distinguished point, whose sign boundary near $\kappa \approx 1.18$ is where the zero-margin center certificate stops transferring.

$\sqrt{1-q_0}$ to eight digits — his zero-margin lemma appears to extend verbatim. On the grid scanned ($\kappa \in \{0, \pm.05, 0.13, \pm.25, \pm.5, \pm.75, \pm1\}$, the interior point on the second lane) the landscape is essentially κ-independent: a single maximum at $(1, 0)$, off-center values below $-3 \cdot 10^{-3}$, negative far rays, Hessian eigenvalues near $(-0.22, -0.0045)$, and moment-body clearance near 0.017. On roughly the half-plane $\lambda_1 < 0$ the s-infimum is $-\infty$: $\alpha\,\mathbb{E}[N^2] = \psi_0$ exactly at the fixed point, so the quadratic tilt coefficient cancels; these directions satisfy the condition trivially and correspond to the capped-tilt case in Huang's treatment.

5 Certificates on a strip

For κ in sub-slabs of width $5 \cdot 10^{-4}$ covering $[-0.65, 0.10]$, with per-slab parameter rectangles seeded by the continuation, we verify in ball arithmetic (`python-flint`, 80-bit working precision):

Two implementation points carry the whole computation. Adaptive complex quadrature cannot resolve enclosure width that comes from parameter balls — subdividing the integration variable does not shrink it, and the integrator exhausts any budget — so every wide-parameter integral is a real-cell Riemann sum. And the ball evaluation of φ/Ψ NaNs on wide arguments in the steep region (the enclosure of Ψ crosses zero), while the ratio is monotone; all Mills-family factors are therefore enclosed by two thin endpoint evaluations, and the sign-critical sums use the mean-value form (midpoint value plus a derivative correction, $O(h^2)$ against the plain hull's $O(h)$). A third point joined below $\kappa = -0.35$: expectations that nearly cancel must be evaluated as one correlated integrand, never as separate sums whose parameter widths add. For the local-concavity check this is the identity (with $t = M'$, $A = 1 - q_0$, and $b = m(\hat{z}) - A$ for $m(\hat{z}) = \mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0}Z))^{-1}]$)

$$\lambda(\hat{z}) = \hat{z} + \alpha\,\mathbb{E}\left[\frac{b\,t^2}{A(A+bt)}\right],$$

whose integrand is monotone in t for fixed (m, A), regular on all of $t \in [0, 1]$, and bounded by $|b|/(Am)$; the separated form's enclosure crossed zero there while the true value stays near -0.14 (the first $[-0.45, -0.35]$ batch, kept as evidence, failed on exactly this). With those in place the checks are:

(i) the contraction $\sup_q (P \circ R_\alpha)'(q) \le \sup_\psi P' \cdot \sup_q |dR/dq| < 1$ on the slab rectangle, via Lemma 2;

(ii) existence: the endpoints of the q interval map strictly inward under $P \circ R_\alpha$ on each of eight α sub-intervals;

(iii) the crossing: at the two α endpoints the fixed point is located by certified Picard iteration and the sign of G resolves through a mean-value form that keeps the stationarity cancellations of Lemma 1 explicit ($G(\alpha_{\mathrm{lb}}) > 0 > G(\alpha_{\mathrm{ub}})$);

(iv) Huang's AMP condition $\alpha\, \mathbb{E}[\mathrm{sech}^4(\sqrt{\psi_0}Z)]\, \mathbb{E}[M'(u(\sqrt{q_0}Z))^2]/(1-q_0)^2 < 1$; and

(v) local concavity at the witness $\hat{z} = -0.6693$: $\lambda(\hat{z}) < 0$.

Together with Lemma 1 (strict decrease of $G_\star$), (i)–(iii) discharge the fixed-point condition verbatim on each slab: $\alpha_\star(\kappa)$ exists, is unique, and lies in the slab's α interval (of width about $4 \cdot 10^{-3}$) for every κ in the slab.

All fifteen hundred slabs covering $[-0.65, 0.10]$ passed every check (forty independent worker runs, contiguous coverage; two first attempts failed and are kept as evidence — one at doubled slab width, since the q pad must also cover the within-slab drift $dq_\star/d\kappa$, which reaches 3.4 per unit κ near -0.2, and one on the separated-form λ enclosure described above). A second lane of one hundred eighty slabs extends the strip to 0.19 (Section 7), with uniqueness supplied by Nakajima's theorem under a certified premise instead of the rectangle contraction, for sixteen hundred eighty in all. Certified point intervals at five margins tighten the slab pads by bisection on the certified sign of G: $\alpha_\star(0.05) \in [0.780449, 0.781325]$, $\alpha_\star(-0.05) \in [0.889115, 0.890013]$, $\alpha_\star(0.0995) \in [0.732844, 0.733711]$, $\alpha_\star(-0.45) \in [1.557652, 1.557655]$, and, past the product bound's reach on the second lane, $\alpha_\star(0.13) \in [0.705932, 0.705934]$, each containing its continuation value (endpoints displayed rounded outward). The certified $\alpha_\star(\kappa)$ intervals have width about $3.5 \cdot 10^{-3}$; the slab at $\kappa \in [0, 5 \cdot 10^{-4}]$ contains the Ding–Sun value $0.833078599\ldots$, and a mid-slab continuation solve lands inside its interval. Thus conditions (i)–(v) — the fixed-point condition verbatim, the AMP condition, and the local-concavity condition — hold for every $\kappa \in [-0.65, 0.19]$ (past 0.10 through the second lane); of Huang's four conditions only the global two-variable bound remains open on the strip. Across the slabs the certified contraction margins run monotonically from 0.069 at $\kappa = -0.05$ through 0.029 at $\kappa = 0.05$ to 0.0083 on the last slab before $\kappa = 0.1$, placing the saturation of the product bound near $\kappa \approx 0.12$ (a projection past the certified range, not a finding); on the negative side the margins grow, and uniqueness there is carried entirely by our own contraction certificates, since Nakajima's theorem is stated for $\kappa \ge 0$.

6 The center of the two-variable landscape at general κ

The first ingredient of the global condition's certificate at $\kappa = 0$ is local maximality of the fixed-tilt functional at the distinguished point, via the Hessian $M = \nabla^2 \bar{\mathcal{S}}_\star(1, 0)$. The second-derivative expansion before the zero-margin specialization gives, at every κ, the same quadratic form $-\langle u, \mathrm{diag}\, I\, u\rangle + C_1 a^2 + C_2 ab + C_3 b^2$ in $a = I_1 u_1 + I_2 u_2$, $b = I_2 u_1 + I_3 u_2$, with the prior-side integrals I_j unchanged and the constraint-side constants C_j now given by integrals at the shifted law (their zero-margin closed forms use $\alpha_\star \mathbb{E}[N\hat{H}] = -q_0(1-q_0)\psi_0$,

which does not survive; $\alpha_\star \mathbb{E}[N^2] = \psi_0$ does). A float transcription of the general-κ constants lands inside every certified interval of the zero-margin appendix (I_j to their printed digits, C_1 through C_3, $\det M$), and along the curve it gives a definite boundary: M stays negative definite up to $\kappa = 1.15$ and loses definiteness between 1.15 and 1.20, while the Hessian of the tilt-optimized functional remains negative out to at least $\kappa = 1.5$. The same formulas evaluated in ball arithmetic on the Ding–Sun rectangle certify $M_{1,1} < 0$ and $\det M > 0$ at $\kappa = 0$ — an independent re-verification of the zero-margin claim through the general-κ route, with the near-cancellations $1 - 4q_0 + 3p_4$ and $q_0 - p_4$ evaluated as single integrands and the constraint-side integrals by mean-value Riemann sums. At $\kappa = 0.05$, deepening the bisection quadrature locates $\alpha_\star$ to $[0.781073068, 0.781074776]$ (width $1.7 \cdot 10^{-6}$), and on the resulting boxes ($3.6 \cdot 10^{-6}$ in q_0, $5.2 \cdot 10^{-5}$ in ψ_0) the certificate resolves: $M_{1,1} < 0$ and $\det M > 0$ in ball arithmetic — the local maximality at the distinguished point, the center ingredient of the two-variable condition, certified at a nonzero margin. The same chain at $\kappa = -0.05$ ($\alpha_\star \in [0.889408032, 0.889409783]$), at the strip's thin-margin edge $\kappa = 0.0995$ ($\alpha_\star \in [0.733478514, 0.733480204]$), and at the deep-negative edge $\kappa = -0.45$ ($\alpha_\star \in [1.557652103, 1.557654133]$, where the constraint-side constants sit far from their zero-margin values, $C_1 = -0.855$ against -0.718) also certifies, and so does $\kappa = 0.13$ on the second lane ($\alpha_\star \in [0.705932218, 0.705933898]$), so the center certificate now holds at six margins spanning the two-lane strip, two of them on the relaxed side where uniqueness has no analytic backup and one past the product bound's reach. The two are consistent by the Schur relation — the fixed-tilt form dominates the optimized one — so the condition itself survives where the fixed-tilt certificate does not: past $\kappa \approx 1.18$ the local argument must re-optimize the tilt, as the zero-margin verification's ray certificates already do.

7 What remains

The one open condition at $\kappa \neq 0$ is the global two-variable bound, whose certified analogue at $\kappa = 0$ is the Region I/II machinery of the capacity verification; its center ingredient (local maximality at the distinguished point) is now certified at four margins spanning the strip, and the scanned landscape suggests the global part transfers with essentially unchanged margins. Past the strip's positive edge the rectangle product bound saturates near $\kappa \approx 0.12$, but the pointwise contraction does not, and the wall is demonstrably a method artifact: at $\kappa = 0.13$ the box product reads 1.32 while a certified sign bisection of the displacement $P(R(q)) - q$ (no contraction input) locates the fixed point at both α endpoints to width $2 \cdot 10^{-8}$, the contraction on the located intervals reads 0.79, and the G crossing resolves with margins $2 \cdot 10^{-3}$ (`past_wall_0p13.py`); Nakajima's hypothesis is certified alongside ($\alpha_{\mathrm{ub}} < \alpha_c(0.13) = 1.0394\ldots$ in ball arithmetic), so with his theorem supplying uniqueness this is the fixed-point condition at that margin, and a strip past 0.12 needs only the existence sweep, which is already contraction-free. That strip is now certified: 180 slabs on $[0.10, 0.19]$, each carrying the premise $\alpha_{\mathrm{ub}} < \alpha_c(\kappa)$ in ball arithmetic, the inward sweep, per-sub-ball sign-bisection location with the contraction consumed only on located intervals (margins ≥ 0.18), and the G crossing. The lower bound is a separate matter entirely: Ding–Sun's conditioned second-moment reduction is proved at $\kappa = 0$ only, so nothing here should be read as an exact threshold statement at nonzero margin — the certified content feeds Huang's upper-bound theorem. On the relaxed side ($\kappa < 0$) the same certificates apply but saddle uniqueness

has no analytic backup, and the approach to replica-symmetry breaking territory at large negative margin deserves its own study.